

ABU DARDA

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Summary

- AI/ML Engineer specializing in Generative AI and Large Language Models (LLMs), with hands-on experience in developing scalable AI solutions using Python and Node.js
- Strong analytical and problem-solving skills, with a proven ability to drive innovation and deliver impactful AI applications.

Skills

Technical Skills

Generative AI & LLMs: RAG systems, Agentic Workflows, Multimodal AI, Fine-tuning LLMs (LoRA, QLoRA, PEFT)

Machine Learning: Deep Learning, NLP, Computer Vision, Data Augmentation.

Cloud & Services: AWS, Azure, GCP.

Software Development: Python, FastAPI, Node.js

Databases: PostgreSQL, MongoDB, GraphQL, Vector Databases (e.g., Pinecone, FAISS)

Tools and Frameworks: TensorFlow, PyTorch, Langchain, Langgraph

Additional skills

Problem Solving | Communication | Teamwork | Leadership | Agile Project Management | Client-Facing Communication

Experience

Brain Station 23

SOFTWARE ENGINEER I

Aug 2024 – Feb 2025

- Led development of **Upendnow**, managing end-to-end implementation to deliver high-quality results on time.
- Managed client interactions to gather requirements, provide updates, and ensure alignment with client expectations.

UpendNow

A Generative AI-powered film-making co-pilot, using LLMs, LoRA fine-tuning, and multimodal AI agents. It creates film scripts and dialogues based on user criteria, generates AI-based scene images and demo videos, and analyzes video content to extract themes and details.

Technologies Used: Serverless, Node.js, AWS, PostgreSQL, React.js

- Fine-tuned Flux image generation models and trained multiple **LoRAs** to achieve scene and character consistency across images.
- Developed a **multimodal AI agent** for video analysis.
- Designed agentic workflows to generate dynamic and context-aware scripts.

ASSOCIATE SOFTWARE ENGINEER

Jan 2024 – Aug 2024

- Built and managed multiple B2B SaaS applications, overseeing the development process.
- Ensured project success by coordinating with stakeholders and delivering high-quality solutions within deadlines.
- Gained hands-on experience in ML model training, evaluation, and deployment.

Chatbot Broilerplate

Boilerplate of chatbots that supports creating different chatbots for different RAGs. Chatbots are customizable according to client needs.

[Try it here](#)

Technologies Used: LLM, FastAPI, AWS, PostgreSQL, Langchain, LangGraph

- Integrated LangChain & LangGraph to implement agentic workflows for dynamic data retrieval, reducing response time by 40%.
- Introduced a multi-turn feature where the chatbot will ask the user for clarification if the query is unclear.
- Implemented chatbot performance evaluation feature, enabling users to assess the quality of generated responses.

Workflow Automation System

A system to classify job descriptions and CVs. It automates document sorting, AI-based reviews, and notifications, simplifying HR processes.

Technologies Used: AWS, Python, FastAPI

- Developed a document classification system with 95% accuracy.
- Automated the whole system, providing quick and accurate insights without human interactions.

BRAC University

TEACHING ASSISTANT

Jan 2023 – Nov 2023

- Assisted course instructors in designing curriculum, conducting classes and grading exams.
- Held consultations with students to clarify concepts and enhance comprehension of course material.

Education

BRAC University, BSc in Computer Science

2020 – 2023

- CGPA: 3.78/4.0
- **Thesis:** An open-source population simulator to provide projections, forecasts, and models of diverse population behaviors under specific stimuli, such as disease outbreaks, natural disasters, and significant congestion.
[Github](#) | [Paper](#)

Achievements

- **Champion**, [R@DIX2.0 - BUCC Week - Programming Contest](#), BRAC University - 2022
- **Champion**, [BRACU Intra-University Programming Contest](#) - 2021
- Reached Specialist in [codeforces](#) and solved more than 1500 problems in [atcoder](#), [codechef](#), and [codingame](#)

Personal Projects

- **Chatbot** - LLM-powered RAG chatbot with multiple knowledge sources and autonomous agents. [github](#)
- **M-link** - A link clustering memetic algorithm for overlapping community detection. (Implementation) [github](#)
- **Facial Expression Detection (CNN)** - Detects emotion from face images [github](#)
- **Glaucoma Detection (CNN)** - Evaluation of different CNN models to detect glaucoma from images. [github](#)
- **Django E-commerce System** - Django web-app to store and sell products. [github](#)
- **Employee Management System** - Java-based employee management system for organizations. [github](#)
- **Interactive quiz Application** - Java-Based web-app to host, create, and manage quizzes. [github](#)
- **Hospital Management System** - Python-based employee management, appointment scheduling and transaction service. [github](#)

Certifications

[Introduction to Generative AI - Google](#)

[TensorFlow-Keras Bootcamp - OpenCV University](#)

[Introduction to Large Language Models - Google](#)